MEANS OF AN EXPONENTIAL PAIR ON ONE PAGE

sec and sech are harmonic means

When two numbers multiply to 1, their harmonic mean is the reciprocal of their arithmetic mean.

THE RULE

Four means of two numbers

The harmonic mean is the reciprocal of the average of the reciprocals, the right mean when reciprocals are what add, as times do over equal distances. For two numbers a and b,

$$\mathrm{AM} = \frac{a+b}{2}, \quad \mathrm{GM} = \sqrt{ab}, \quad \mathrm{HM} = \frac{2}{\frac{1}{a}+\frac{1}{b}} = \frac{2ab}{a+b}, \quad \mathrm{RMS} = \sqrt{\frac{a^2+b^2}{2}},$$

and the third is tied to the first two by

$$\mathrm{AM} \times \mathrm{HM} = \mathrm{GM}^2, \qquad \text{so} \qquad ab = 1 \implies \mathrm{HM} = \frac{1}{\mathrm{AM}}$$

Both exponential pairs, $\mathrm{e}^{x}, \mathrm{e}^{-x}$ and $\mathrm{e}^{\mathrm{i}x}, \mathrm{e}^{-\mathrm{i}x}$, have product 1, so their geometric mean is 1 and their harmonic mean is the reciprocal of $\cosh x$ or of $\cos x$.

speeds 30 and 60 over equal distances

RMS	47.4
AM	45
GM	42.4
HM	40

The true average speed is the harmonic mean, 40; always HM ≤ GM ≤ AM ≤ RMS for positive numbers.

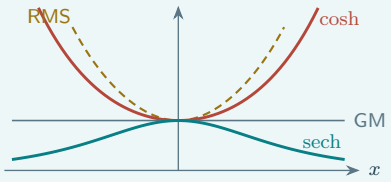
THE REAL PAIR $\mathrm{e}^{x}, \mathrm{e}^{-x}$

Hyperbolic: the order holds

$$\mathrm{AM} = \cosh x, \quad \mathrm{GM} = 1, \quad \mathrm{HM} = \operatorname{sech} x, \quad \mathrm{RMS} = \sqrt{\cosh 2x}$$

Both numbers are positive, so the chain holds, with equality only at $x = 0$:

$$\operatorname{sech} x \le 1 \le \cosh x \le \sqrt{\cosh 2x}.$$



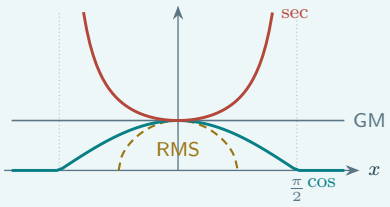
THE IMAGINARY PAIR $\mathrm{e}^{\mathrm{i}x}, \mathrm{e}^{-\mathrm{i}x}$

Circular: the same formulas, a new order

$$\mathrm{AM} = \cos x, \quad \mathrm{GM} = 1, \quad \mathrm{HM} = \sec x, \quad \mathrm{RMS} = \sqrt{\cos 2x}$$

The pair is not positive, so on $|x| < \frac{\pi}{2}$ the harmonic and arithmetic means swap places, and the RMS is real only for $|x| \le \frac{\pi}{4}$:

$$\sqrt{\cos 2x} \le \cos x \le 1 \le \sec x.$$



THE IDENTITIES BEHIND THE MEANS

The Pythagorean identities are statements about means

Two exact relations hold for any pair: $\mathrm{AM}^2 - \mathrm{GM}^2 = \left(\frac{a-b}{2}\right)^2$ and $\mathrm{RMS}^2 = \mathrm{AM}^2 + \left(\frac{a-b}{2}\right)^2$. The half-difference of the real pair is $\sinh x$ and of the imaginary pair is $\mathrm{i}\sin x$, so the two relations become the familiar identities.

	$\mathrm{e}^{x}, \mathrm{e}^{-x}$	$\mathrm{e}^{\mathrm{i}x}, \mathrm{e}^{-\mathrm{i}x}$
half-difference $\frac{a-b}{2}$	$\sinh x$	$\mathrm{i}\sin x$
$\mathrm{AM}^2 - \mathrm{GM}^2 = \left(\frac{a-b}{2}\right)^2$	$\cosh^2 x - 1 = \sinh^2 x$	$\cos^2 x - 1 = -\sin^2 x$
$\mathrm{RMS}^2 = \mathrm{AM}^2 + \left(\frac{a-b}{2}\right)^2$	$\cosh 2x = \cosh^2 x + \sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x$
$\mathrm{AM} \times \mathrm{HM} = \mathrm{GM}^2$	$\cosh x \operatorname{sech} x = 1$	$\cos x \sec x = 1$

A BRIDGE BETWEEN THE PAIRS

The Gudermannian swaps the roles

The angle $\operatorname{gd} x = \arctan(\sinh x)$ gives $\sec(\operatorname{gd} x) = \cosh x$ and $\cos(\operatorname{gd} x) = \operatorname{sech} x$, so on this angle the circular harmonic mean equals the hyperbolic arithmetic mean. Its inverse, $\int_0^{\varphi} \sec t \, \mathrm{d}t = \ln(\sec\varphi + \tan\varphi)$, is Mercator's spacing of latitudes.

WHERE THE HARMONIC MEAN APPEARS

The shape of a solitary wave

A shallow-water soliton has the profile $A \operatorname{sech}^2\left(\frac{x-ct}{w}\right)$ and an optical-fibre pulse keeps the shape sech, both built from the harmonic mean of e^{x} and e^{-x}; and $\frac{\mathrm{d}}{\mathrm{d}x}\tanh x = \operatorname{sech}^2 x$ gives the logistic curve its bell-shaped growth rate.

Conventions. $\sec = 1/\cos$, $\operatorname{sech} = 1/\cosh$ and gd is the Gudermannian; the means extend to complex numbers as formulas, but HM ≤ GM ≤ AM ≤ RMS

is guaranteed only for positive numbers.